

Chapter 8: Statistics & Probability

Central Tendency, Dispersion, Combinatorics & Conditional Probability

8.1 Descriptive Statistics & Measures of Spread

Statistics analyzes numerical datasets through central tendency (Mean, Median, Mode) and dispersion (Variance, Standard Deviation).

Mean & Variance

- Mean (μ): $\sum x_i / N$
- Variance (σ^2): $\sum (x_i - \mu)^2 / N$
- Standard Deviation (σ): $\sqrt{\sigma^2}$

Combinatorics

- Permutations (Order matters):
 $P(n, r) = n! / (n - r)!$
- Combinations (Order irrelevant):
 $C(n, r) = n! / [r!(n - r)!]$

Worked Example 8.1: Standard Deviation Calculation

Problem: Calculate the mean and variance for data set: {2, 4, 6, 8}.

Solution:

1. Mean $\mu = (2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$.
2. Squared deviations: $(2-5)^2=9$, $(4-5)^2=1$, $(6-5)^2=1$, $(8-5)^2=9$.
3. Variance $\sigma^2 = (9 + 1 + 1 + 9) / 4 = 20 / 4 = 5$. Standard Deviation $\sigma = \sqrt{5} \approx 2.24$.

8.2 Probability Rules & Conditional Probability

Probability measures the likelihood of an event E : $0 \leq P(E) \leq 1$.

- Addition Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Conditional Probability: $P(A|B) = P(A \cap B) / P(B)$
- Independent Events: $P(A \cap B) = P(A) \cdot P(B)$
- Bayes' Theorem: $P(A|B) = [P(B|A) \cdot P(A)] / P(B)$

Worked Example 8.2: Conditional Probability

Problem: A card is drawn from a standard 52-card deck. What is the probability that it is a King, given that it is a face card?

Solution:

Let A = King (4 cards), B = Face card (12 cards).

$P(A \cap B) = 4/52$, $P(B) = 12/52$.

$P(A|B) = (4/52) / (12/52) = 4/12 = 1/3$.

Chapter 8 Review & Practice Problems

1. Find the median and mode of the data set: {12, 15, 12, 18, 22, 15, 30, 15}.
2. In how many ways can a committee of 4 students be selected from a class of 10?
3. Two fair dice are rolled. Find the probability that the sum of the numbers is 8 or greater.
4. Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$. Are A and B independent? Find $P(A \cup B)$.

